SAHIL KUREEL

Senior Software Engineer | Backend & Distributed Systems

Atal Bihari Vajpayee - Indian Institute of Information Technology & Management, Gwalior - (B.TECH + M.TECH) In Information Technology
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Backend Software Engineer with 5+ years of experience building scalable APIs, distributed systems, payment platforms, and cloud-native applications. Experienced in asynchronous processing, API design, production troubleshooting, performance optimization, CI/CD, and system reliability. Familiar with LLM integrations and Retrieval-Augmented Generation (RAG) applications.

Skills

Programming languages : Python | Java 17+ | JavaScript (ES6+) | SQL | C / C++ | Bash
Frameworks : FastAPI | spring-boot | Django | NodeJs | ReactJs | ExpressJs | Maven
Storage systems :
- PostgreSQL / AuroraDB (RDS) | DynamoDB | S3 / GCS / Firehose | MongoDB | Redis / Memcached

Cloud Tech & Tools :
- AWS | GCP | LocalStack | OpenTofu | Terraform | Docker | Jenkins | SonarQube | Git
- Linux Systems | Distributed Systems | System Designs at High Level as well as Low Level.
- Performance Optimization & Tuning for scale | Data Structures & Algorithms | REST API Designs, SOLID Principle
- Concurrency Control using Mutex, Semaphores for Multithreading and Asynchronous Programming
- Encryption at REST using AES-CBC | AES-GCM

AI & Data Tools :
- Generative AI / Retrieval-Augmented Generation (RAG)
- DynamoDB-Streams / PySpark / Apache-Airflow
- NumPy / Pandas / SciPy / scikitlearn
- LLMs integration with Ollama / Llama.cpp with GPT-5 / Claude Haiku 4.5, FastAPI as serving interface

Experiences

Mar'26 - Sept'26 | **HandShake AI**
AI Implementation Engineer · Bengaluru
AI Coding-Agent Evaluation and RAG Platform | Project Dynamo, Terminal Bench 2, Sentinel Ultra
Tech: RAG, LLMs, Python3, FastAPI, Django, NodeJs/ExpressJs, Java/Spring Boot, DynamoDB, RDS, Docker, ReactJs, Tailwind CSS.
Project Dynamo, an AI engineering initiative for creating repository-level tasks used in **Terminal-Bench-2** to evaluate coding agents in realistic terminal environments.
Insights: Built and maintained benchmark-oriented software projects spanning Python/FastAPI/Django, Node.js/Express and Java/Spring Boot, including repository scaffolding, application setup, tooling and test workflows. Developed AI-assisted workflows involving RAG applications and LLM-based engineering tasks, enabling AI systems to retrieve project context and perform multi-step software-engineering operations.

Aug'21 – Sept'25 | **HEAD DIGITAL WORKS PVT. LTD.**
SDE - III (Platform Team) · Bengaluru
Unified Payment Platform | Distributed Payment Processing
Tech: Java(spring-boot), AWS - ECS, Lambda, API Gateway, DynamoDB, RDS, Kafka, Docker
Scale: 10M transactions/day, 100K TPM
Impact: 10M transactions/day and 100K TPM, with an architecture designed for horizontal scaling and third party provider-independent integration.
Insights: Built a unified abstraction over payment aggregators such as Juspay and Razorpay, while also supporting KYC and bank-account validation. The key engineering challenge was making the platform reliable despite external payment-provider latency, retries and asynchronous callbacks. Separated synchronous API operations from asynchronous processing using RabbitMQ/Lambda and used DynamoDB/event streams for scalable event processing. Owned production reliability and performance issues, rather than only implementing individual APIs.

Distributed Financial Reconciliation Engine | Financial Infrastructure
Tech: Python3(Django), Java(Maven), AWS - DynamoDB Streams, Lambda/ECS, RabbitMQ, PostgreSQL (AuroraDB-AWS), Docker.
Scale: 50K+ transactions/day.
Impact: 99.5% financial discrepancy detection, 15+ hours/week manual effort saved.
Insights: Designed a distributed reconciliation engine that continuously consumed transaction changes and compared records across multiple ledger systems. Instead of doing reconciliation synchronously, you used DynamoDB Streams + Lambda + asynchronous workers to distribute the workload. You added concurrency control, retries, scheduled jobs, audit reporting and transaction-status tracking. The system automatically flagged 99.5% of discrepancies and removed more than 15 hours of weekly manual reconciliation work.

Aug'20 – July'21 | **GARG CONSULTANCY CO.**
Software Engineer - Backend · Gurugram
Tech: Python(Flask/Django), NodeJs(ExpressJs), AWS, PostgreSQL, RabbitMQ, Docker
Insights:
- Led backend development for financial and digital platforms using Python and NodeJs.
- Designed and implemented RESTful APIs, CMS systems and backend services with scalability and security best practices.
- Collaborated with product and clients to deliver production-ready solutions.
- Mentored junior developers and improved engineering workflows.
- Managed cloud deployments and backend infrastructure on AWS.
- Supported frontend integrations with ReactJS & NextJs applications.

Certificates & Grants

2017 Management Team (Technical Assistant) at ACM - ICPC, Gwalior region.
2019 AWS Certified Solution Architect Associate.